TEAM PEAKYGLIDERS · GURU NANAK DEV UNIVERSITY

SYNERA

Trusted AI for Multi-Stressor Environmental Intelligence

Discover. Validate. Verify. Act.

Study report for the team — problem, workflow, real vs simulated data, accuracy, cybersecurity and ledger, full tech stack, and what is left.
Every number in this report was read from the project's own result files and README (computed by the code), not invented. Where something is our own judgement, it says so.
Written in plain language so that every teammate can explain any part of the project to a judge.

How to use this report

1. The problem we solve, and why

2. Workflow, page by page

3. Real vs simulated data, limitations, accuracy

4. Cybersecurity and the ledger

5. Tech stack, models, formulas, and why

6. What is left

A. Judge questions and answers

B. Numbers cheat-sheet

The project in one paragraph

Heat, air pollution, drought, humidity, wind and radiation do not hit us one at a time, but most analysis studies them one at a time. SYNERA is a web platform with a Python science engine behind it. An **AI engine hunts** for combinations of environmental stressors that act together (*interactions*). A **strict statistics layer decides** which of those hunches survive. A **cybersecurity and ledger layer proves** that the data, code and results were not tampered with and can be reproduced. A **Scenario Lab** lets a user choose a US state and season, set any mix of stressors, and see the estimated effect on deaths, learned from real CDC, EPA, ERA5 and NASA data, with an honest range.

The five words you must know

Word	Plain meaning
Stressor	An environmental condition that can stress people or nature: heat, PM2.5 smoke, ozone, drought, humidity, wind, rain, UV, radiation.
Interaction	Two stressors together do something different from adding their separate effects. Example: heat + dirty air is worse than heat's harm plus dirty air's harm.
False discovery	A "finding" that is only luck. Test enough combinations and some will look real by chance. Most of our engine is built to stop this.
Provenance	The recorded history of a result: which data, which code, which settings, when. Our ledger stores it so it cannot be quietly changed.
Hash (SHA-256)	A 64-character fingerprint of a file. Change one bit and the fingerprint changes completely. Two files with the same fingerprint are identical.

COLOUR TAGS USED IN THIS REPORT

REAL measured from real-world records SIMULATED generated by us with a known answer

MIXED real inputs, simulated outcome

1 The problem, the innovation, and why we chose it

1.1 Problem statement (written for what the project is today)

> **PROBLEM STATEMENT**
>
> Climate and environmental harm usually comes from **several stressors acting together** — for example a heatwave during smoky, polluted air. Decision-makers need to know which combinations are really dangerous, but (1) analysis almost always studies one stressor at a time, (2) when AI or statistics searches many combinations at once it produces **convincing-looking false discoveries**, and (3) the pipeline from raw data to a published number is **easy to corrupt and hard to audit**, so nobody can be sure the result is trustworthy. We built **SYNERA**: a platform that discovers possible combined effects with AI, validates them with strict statistics, makes the whole computation tamper-evident and reproducible, and turns what survives into decision support — including an interactive **Scenario Lab** grounded in real health outcomes.

1.2 What exactly are we trying to solve?

Three connected problems. Each one is real, and each one is addressed by a different part of SYNERA.

Problem	Why it matters	What answers it in SYNERA
1. Compound risk is under-studied.	Heat, smoke, drought and humidity arrive together (wildfire season, monsoon heat, winter smog). A city planner or a health department needs "what happens when these combine?" Single-variable studies cannot answer that.	Discovery Lab (tests all 45 stressor pairs), Scenario Lab (combine stressors yourself), Risk Brief.
2. AI finds patterns that are not real.	With 10 stressors there are 45 pairs; with a gene study there are 24,688 genes. In a benchmark, a plain pairwise screen was **about 92% false** (92.0–92.8% false discoveries at every noise level). A wrong "finding" that guides climate action wastes money and trust.	The validation stack: FDR control, conditional selection, effect-size gate, confidence interval, second opinions.
3. Nobody can audit the pipeline.	Data can be edited, a model file swapped, code changed after the fact. A reader cannot tell. This is the "reproducibility crisis" in miniature, and it matters more once AI is in the loop.	Provenance ledger (SHA-256, Merkle proofs, Ed25519 signatures), replay check, tamper simulator, Trust Center.

1.3 The real problem we cater to

We cater to people who must **act** on environmental evidence: municipal climate teams, public-health departments, researchers, and anyone reading an "AI says X" claim. What they need is not another number — it is a number they can **trust**, with a clear statement of how sure it is and where it should not be used.

SYNERA's tagline says it: *AI generates hypotheses. Science validates them. The ledger does not prove the science; it makes the computational provenance tamper-evident.*

1.4 The innovation (what is new or unusual)

1. **AI proposes, statistics disposes.** Three different methods (a statistical model, LightGBM with SHAP, and a small neural network) look at the same pairs. A pair is promoted only if it survives strict statistics *and* a second method agrees. Most projects show only the AI's output.
2. **Honest negatives are a feature.** The platform openly says "no interaction survived" when that is true — on real NASA gene data and on real US death data. A finding that disappears when you remove an obvious confounder (we saw ozone × UV vanish once nationwide flu waves were removed) is withdrawn, not reported.
3. **A trust layer around the science.** Every dataset, run and finding is fingerprinted and chained in a signed, append-only ledger. A built-in **tamper simulator** lets a judge attack the data and watch the result get blocked.
4. **An evidence ladder, not a yes/no.** CANDIDATE → VALIDATED → VERIFIED, plus BLOCKED and REJECTED. Eight gates (G1–G8) are evaluated live. Findings built on simulated data are **capped** at VALIDATED by rule.
5. **A measured report card.** We tested our own engine on simulated data where the truth is known (50 random seeds × 4 noise levels, 100 null simulations, 600 security trials) and show where it fails, not only where it works.
6. **Scenario Lab on real data, evaluated in the browser.** The prediction model is a short list of numbers learned from real deaths, so it updates in tens of milliseconds, works even on the free static hosting, and every prediction splits into "separate effects" and "interaction effect".
7. **Runs free for everyone.** A recorded-demo build needs no server; the live version runs locally or on Docker.

1.5 What motivated the problem choice

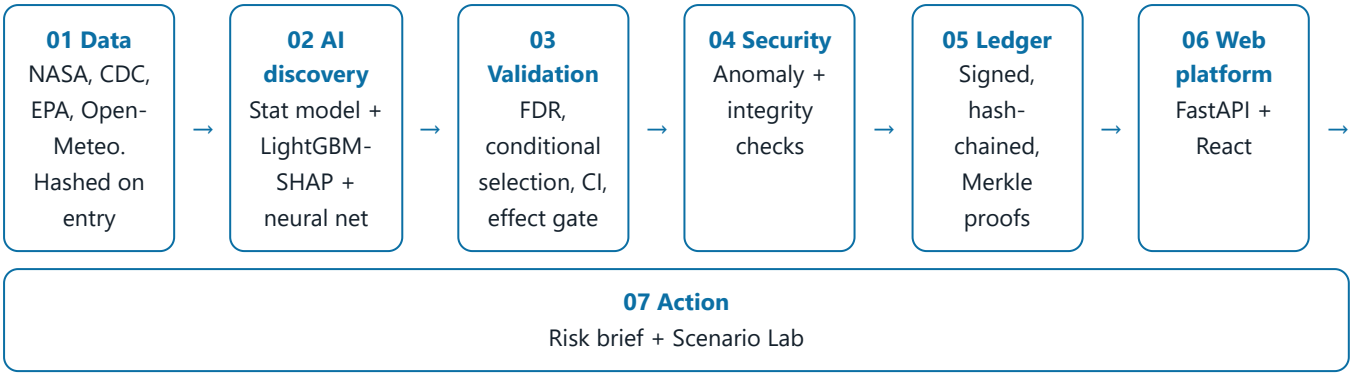
- **The hackathon theme: environment, climate and action.** Compound events (heat + smoke, drought + heat) are exactly where climate impacts are growing and where single-factor thinking fails. "Action" requires trustworthy evidence.
- **AI hype versus AI trust.** Anyone can train a model and see "interesting" patterns. The hard, valuable part is knowing which patterns are real — so we built the part that is usually missing.
- **The reproducibility problem.** Studies that cannot be reproduced, or whose data were altered, damage public trust in climate science. A tamper-evident record is a direct, practical answer.
- **Real data exists but is scattered.** NASA, CDC, EPA and Open-Meteo all publish open data. Joining them honestly (and saying what the join cannot show) is itself useful.
- **Local relevance.** Our eight benchmark cities include Ludhiana and Delhi, where heat and air pollution genuinely overlap.

TEAM TO ADD

Add one or two sentences of your own personal story here (a family member affected by heat or smog, a local event, a class project). Judges remember a personal reason more than a technical one. This report does not invent one for you.

2 Workflow and features, in order

2.1 The big picture (architecture)



The ledger can also "replay": it re-runs a past analysis and checks that it produces the same answer (dotted arrow in the project README).

The three layers of the software

Layer	What lives there	Folder
Science core (Python)	Data loading, discovery models, validation, security checks, ledger, pipeline orchestrator, scenario model, risk assessment. No web code in it.	`src/synera_core`
API (FastAPI)	Turns the science core into web endpoints: datasets, runs (with live progress), findings, security, ledger, benchmark, space biology, risk, scenario. Stores runs/findings in a small SQLite database.	`backend/app`
Website (React)	Ten pages, charts, a 3D globe, guided tour. Every number comes from the API (or from recorded snapshots in the free demo).	`frontend/src`

Two ways the site runs

- **Live mode** (`.\dev.cmd` , Docker): the website talks to the real API, so you can start new analysis runs, launch attacks and re-verify the ledger.
- **Recorded demo** (Vercel): a script runs the real API once and saves every response as a file. The hosted site replays those real recorded outputs with no server. It cannot compute new runs, and the page says so in a banner.

2.2 The data flow of one analysis run (nine steps)

When you press "Run engine", the orchestrator executes nine real steps. Each reports its true duration; there are no fake delays.

#	Step	What happens

1	**Load data**	The dataset's SHA-256 is compared with the one registered in the ledger. A mismatch stops the run: BLOCKED.
2	**Security pre-check**	Scans for impossible values, duplicates, outliers, drift, flipped labels, odd time order.
3	**Discovery**	Tests every stressor pair with a panel model, then asks LightGBM+SHAP and the neural net for their own rankings.
4	**Validation**	False-discovery control (BH-FDR), conditional selection, bootstrap confidence intervals, minimum-effect gate.
5	**Replication**	Re-tests accepted pairs on two disjoint halves of the cities (Track B) or compares the second tissue (Track A).
6	**Provenance**	Builds a run manifest (data hashes, code hash, settings, seed) and appends an EXPERIMENT_RUN block to the ledger.
7	**Security post-check**	Re-checks ledger chain, dataset hash and whether the source code changed during the run.
8	**Reproducibility replay**	Runs the whole analysis again from the same seed; the digest of the results must be identical.
9	**Finalize**	Evaluates the eight gates, assigns each finding a status, writes ledger blocks, stores findings.

2.3 Every page, in the order a visitor sees them

Page 1 — Landing page "Observatory" (`/`)

What it shows. A NASA-style Earth observatory home: an Earth view (a 3D globe where the device supports it, a static image otherwise) with animated stressor beams, the headline message ("The environment doesn't change one variable at a time"), and a live strip of counts taken straight from the engine — how many findings are candidates, validated, verified and blocked, including honest zeros. It has buttons for the guided tour and a call-to-action into the Scenario Lab, plus a "Track C · real outcomes" card.

Why it exists. A judge decides in the first ten seconds whether to keep listening. The page states the problem, proves the numbers are live, and routes to the strongest feature. Showing zero verified findings on the home page is deliberate — it signals honesty.

Page 2 — Guided tour (3 minutes, 8 stops)

What it does. A floating card walks through the pages in deck order (problem → discover → validate → verify → space biology → Scenario Lab → benchmark → act) and navigates for you. The card is translucent and blurred so it reads on any background. **Why:** it lets a judge see the whole story without you speaking, and gives you a safe script if the demo is rushed.

Page 3 — Discovery Lab (`/discovery`) — step 01 DISCOVER

What it does. You pick a dataset and press "Run engine". You watch the nine steps with real timings. Then you see:

- **What survived**: a funnel of counts, from all 45 pairs down to the final findings.
- **Interaction map**: a 10×10 matrix of how strong each pair looks.
- **Response surface**: a contour plot of the outcome across two stressors. The dashed contours show "what you would expect if effects simply added up"; the solid contours show what the model fitted. The **gap between them is the interaction**.
- **Three kinds of interaction**: synergistic (worse together), antagonistic (they cancel), non-linear (a threshold or curve).
- An explainer, "Why pairs get explained away": a pair can look significant just because it resembles a truly interacting pair; conditional selection removes those look-alikes.

Why it matters. This is the heart of the idea: finding combined effects. It is also where a judge can see that the work is real (a live run, not a screenshot).

Page 4 — Findings registry (`/findings`) — step 02 VALIDATE

What it does. A searchable, filterable table of every finding with its status chip (CANDIDATE, VALIDATED, VERIFIED, BLOCKED, REJECTED), its data provenance (REAL or SIMULATED), effect size and confidence score. **Why:** it turns raw statistics into an auditable list and enforces the rule that simulated findings are visibly labelled and capped.

Page 5 — Evidence dossier (`/findings/:id`)

What it does. One finding in full, as nine labelled panels: A response surface, B effect estimates, C replication, D method agreement (three methods), E statistics, F trust stack (the eight gates, each with a pass/fail and the reason), G fingerprints and ledger (hashes and Merkle proof), H run receipt (everything needed to reproduce it), I history. **Why:** a reader can see exactly why a finding earned its status — no black box.

Page 6 — Trust Center (`/trust`) — step 03 VERIFY

What it does. Explains the vulnerable path from raw data to a published result, then the SYNERA trust layer. It contains the **tamper simulator**: choose an attack (bit flip, subtle edit, row edit, outlier injection, unit drift, label poisoning, model swap, ledger tamper), press "Launch attack", and see which defences notice. The attack runs on a temporary copy; your real data and ledger are never touched. You can then re-verify the real ledger chain: still valid. **Why:** security you can watch is more convincing than security you describe. Details are in section 4.

Page 7 — Space Biology (`/space-biology`) — the real-data testbed

What it does. Re-analyses real NASA GeneLab mouse studies: chronic low-dose radiation × hindlimb unloading (a ground model of microgravity). It tests 24,688 genes for a radiation × unloading interaction. Sections: the experiment design (6 control / 6 radiation / 3 unloading / 6 both), counts of interactions found (zero in each class), statistical power, the exploratory top-20 genes (clearly marked as not findings), cross-tissue agreement, and quality control (null calibration, positive controls).

Why it matters. It is the project's real-data proof that the engine does not invent discoveries: with 21 animals, nothing survives, and the page says so — and tells you the study would need an effect near 4 log2 to be 80% likely to detect it. A tool that returns "nothing" on underpowered data is a tool you can trust when it does return something.

Page 8 — Scenario Lab (`/scenario`) — "Your turn"

What it does.

1. **Where and when:** choose an outcome (all deaths, heart disease, or stroke), a US state and a month.
2. **Set the conditions:** ten sliders (heat, PM2.5, humidity, ozone, drought, PM10, NO2, UV, wind, rain) in real units. Presets: "Heatwave", "Smoky air", "Heat + dirty air", "Clean-air week", and so on.
3. **Estimated consequence:** the extra (or fewer) deaths versus a normal week in that state, in percent, with a 90% range, plus an "extrapolation" warning if the setting is far beyond what that state has ever experienced.
4. **Where it comes from:** splits the answer into separate effects added up, and extra from interaction; shows what cleaner air would change.
5. **Every weight of every stressor:** a curve per stressor — you see the consequence at every dose, not only the one you picked.
6. **Two stressors at once:** a 2D map of two stressors together.
7. **Trust panel:** the real findings (separate effects), the check on states the model never saw, the interaction check, and a "read this before using a number" box.

Why it matters. This answers "what happens if…?" with **real** outcomes. It is the feature a non-expert can play with, and it shows calibration and limits instead of pretending to be a crystal ball. The model is evaluated in the browser; a test confirms the browser version matches the Python version to 9 decimal places.

Page 9 — Benchmark / Engine report card (`/benchmark`) — "How good is it?"

What it does. Seven sections: headline, where false discoveries die (stage by stage), null p-value histograms (flat means honest), whether variance moderation helps with few samples, what gets found and what the effect gate costs, whether the security layer catches tampering, and method notes. A red box at the top says what the report does **not** show.

Why it matters. This is the proof of the claim "our engine controls false discoveries". It is measured on simulated data with a known truth, and includes the places we are weak.

Page 10 — Risk Brief (`/risk`) — step 04 ACT

What it does. For one of the eight cities, it fetches the current and 7-day forecast (live from Open-Meteo, with a cached fallback), converts it to the same standardised units the findings were fitted in, and evaluates each finding's response surface at the forecast values. It reports the additive expectation, the modelled response, and the **interaction uplift**, then a tier (LOW, MODERATE, HIGH, SEVERE) with an uncertainty band, the source finding ids and a reproducibility receipt.

Why it matters. This is the "Action" in the theme: validated findings become a brief a city could use. **Important honesty:** because today's findings come from simulated data, every brief is labelled a **simulated demonstration**; the system never shows a tier without its band and sources.

Page 11 — Guide (`/guide`) — "How SYNERA works"

The seven layers, the life of a finding, where the data comes from, a glossary and source attributions. Hover-tooltips on technical terms across the site use the same glossary. **Why:** judges and non-experts can learn

the vocabulary without leaving the product.

2.4 The life of a finding

Status	Rule
CANDIDATE	Survived false-discovery control *and* conditional selection (it is not just a look-alike of another pair).
VALIDATED	Effect at least the pre-set minimum (0.05 of the outcome's standard deviation), a bootstrap confidence interval that clears it, and at least one second method agrees (robustness at least 2 of 3).
VERIFIED	Validated *and* replicated in independent data, *and* built from REAL data, *and* dataset hash and ledger chain intact, *and* the run replays to the same result.
BLOCKED	Any integrity check failed (data hash, ledger chain, ledger inclusion, replay or code change, blocking security check). "Result blocked from verified status."
REJECTED	Detectable but smaller than the minimum effect, so not worth acting on.

> **SAY THIS CAREFULLY**
>
> VERIFIED means "statistically validated with verified computational provenance". It never means "biologically proven". Today the count is **3 VALIDATED (simulated, capped), 0 VERIFIED, 0 BLOCKED**.

2.5 Backend API in one view

All under `/api/v1`: `system` (health), `datasets` (list and verify hashes), `runs` (start, list, live progress stream), `findings` (list, detail, gates), `interactions` (matrix, surface), `security` (tamper simulation, ledger verify), `space-biology`, `benchmark`, `risk`, and `scenario`. Live progress uses **Server-Sent Events**, a simple one-way stream from server to browser, so the run page updates step by step.

3 Real vs simulated data, limitations, and accuracy

3.1 What is real and what is simulated

Part of the project	Status	What it actually uses
Track A — Space Biology	REAL	NASA OSDR / GeneLab studies OSD-211 (spleen) and OSD-237 (skin), mouse RNA-seq; 21 animals each; 24,688 and 25,716 genes. The two studies share 20 of 21 animals.
Track B — environmental benchmark	MIXED	Inputs are real: 8 cities × 518 days of weather and air-quality (Open-Meteo archives). The **outcome is simulated** with three known planted interactions so we can score the engine against the truth.
Track C — real health outcome	REAL	CDC NCHS weekly deaths (all causes, heart disease, stroke) for 38–46 US states, 2014–2019, joined to EPA Air Quality System monitors, ERA5 weather (Open-Meteo) and NASA POWER UV index. 14,065 complete state-weeks for all-cause deaths over 46 states. COVID years excluded on purpose.
Scenario Lab	REAL	The model is learned from the Track C panels. The sliders are your inputs; the output is the model's estimate.
Risk Brief	MIXED	The forecast is real (live Open-Meteo). The interaction findings it applies are simulated, so the brief is labelled a demonstration.
Findings registry	MIXED	Each finding carries its own provenance tag. The three current VALIDATED findings are from simulated data.
Benchmark report	SIMULATED	Simulated by design: it is the exam we give our own engine.
Security tests	SIMULATED attacks	Attacks (bit flips, edits, etc.) are applied to temporary copies of data; the detection is real code measured on them.
Ledger, hashes, signatures	REAL crypto	Real SHA-256, Merkle trees and Ed25519 signatures over real files and results. Nothing is mocked.
Reference data	REAL	Natural Earth land polygons, gene symbol map (MyGene.info), NASA Visible Earth imagery.

How the real outcome was built (Track C)

1. Weekly deaths per state come from CDC. The **outcome** is "excess mortality in %": observed deaths compared with the state's own expected level for that week (trend and season), **minus the national figure for the same week**.
2. Subtracting the national same-week figure is the key trick. It removes nationwide flu waves and big weather events, so a local stressor cannot "take credit" for them.
3. Each stressor is turned into an **anomaly**: how far today's value is from that state's normal for that week, in units of the state's usual deviation ("+2" = two usual deviations hotter or dirtier than normal there).

4. Raw downloads are cached and not committed; the processed tables are tracked and hashed in the manifest and ledger.

3.2 What the real data found

REAL FINDINGS (TRACK C, ALL-CAUSE DEATHS, 2014–2019)

- **No interaction survives.** 45 stressor pairs × 3 causes of death: none passed FDR control and conditional selection. An earlier pass flagged ozone × UV and heat × drought; both vanished once flu and weather waves were removed.
- **PM2.5 +10 µg/m³ → +0.62%** more deaths that week (90% range 0.34 to 0.83).
- **NO2 +10 µg/m³ → +0.36%** (0.10 to 0.55). **Ozone +10 µg/m³ → +0.09%** (0.03 to 0.16).
- **Heat index 5 °C above normal in summer → +0.36%** (0.06 to 0.68). **5 °C below normal in winter → +0.51%** (0.36 to 0.71). Heat in winter is protective (−0.41%): the same variable does opposite things in different seasons, which is why the model has a heat × season term.
- **Heart disease** responds more strongly: PM2.5 +1.45% per +10 µg/m³ (0.95 to 2.01).
- PM10, humidity, drought, UV, wind and rain showed no clear effect on their own.

These are small percentages — which is realistic. A change of 0.6% in weekly deaths in a state is a lot of people over a population, but tiny next to flu, ageing and chance.

3.3 Limitations: what we faced, what is solved, what remains

Solved or reduced (problems that were found and fixed)

Problem	How it was solved
Daily weather data are autocorrelated, so a normal test says "significant" far too often (inflation λ = 4.11 for the naive test).	Block-bootstrap-calibrated test; λ falls to **1.38** (7.6% of null p-values below 0.05 versus the ideal 5%).
Pairwise screening found ~2,000 look-alike pairs per 150 datasets (92% false).	Conditional selection (forward + backward) cuts strict false discoveries to 0–6.2% depending on noise.
Early security detectors raised alarms on 100% of clean data.	Baseline-calibrated thresholds with a binomial rule; clean-data false alarms now 2–5% per detector, 10% for any warning; statistical detectors are advisory only.
The first orchestrator used canned messages and fixed delays; "second opinions" returned hand-set values; the replay returned "match" whenever the source hash matched.	Rebuilt: nine real steps; real LightGBM+SHAP and PyTorch NID; replay re-runs the analysis and compares result digests.
The demo had no real health outcome.	Added Track C (CDC, EPA, ERA5, NASA POWER) and the Scenario Lab.

A spurious ozone × UV "discovery" appeared on real deaths.	Same-week national adjustment removed it. The engine's negative result is now reported honestly.
Ridge shrinkage halved the real effect sizes.	Penalty split: light on main effects, heavy on pairwise products, chosen by cross-validation (30 and 300).
Hard-coded or fabricated values in the old API and UI (fake groups, fixed risk tier, fake study).	Every endpoint computes from data; tests prevent regression.
The tamper simulator crashed on integer columns and could report "no tampering" for a no-op edit.	Fixed; tests cover boolean, integer columns and zero cells.
Free hosting cannot compute heavy runs.	Recorded-demo build on Vercel plus a low-memory guard on Render.

Still open (honest list)

#	Limitation	Why it remains / how big it is
1	**No finding is VERIFIED.**	VERIFIED needs real data *and* independent replication. Our only real-data outputs are separate effects (not interactions), and no second independent real dataset has been used.
2	**Track B outcome is simulated.**	It tests the engine, not nature. The three VALIDATED findings are capped by rule.
3	**Track A is underpowered.**	21 animals; 6/6/3/6 per group. Only huge interactions (about 4 log2 for 80% power) could be seen. Spleen and skin share animals, so skin is not independent replication.
4	**Track C exposure is crude.**	One metro-area weather location per state; state-mean monitor readings (many monitors sample 1 day in 3 or 6). This blurs real effects toward zero, so true effects are probably somewhat larger.
5	**Ecological, not individual.**	The model says what happens to a state's weekly deaths on average, not an individual's risk, and not proof of cause.
6	**Scenario model explains little.**	Only about 0.7% of week-to-week variation in out-of-sample states. Flu, ageing and chance dominate. A gradient-boosting model does no better.
7	**Non-linear shapes are hard to type at low noise.**	The engine names the surface correctly only 26% (SNR 1) and 44% (SNR 2) of the time; 90–100% from SNR 5 up.
8	**The calibrated test is slightly anti-conservative.**	7.6% of null p-values under 0.05 (ideal 5%). HAC is reported only as a sensitivity check.
9	**Statistical security detectors are weak.**	MCD and the autoencoder never fired on the tested attacks yet raise ~5% false alarms; subtle in-range edits and label flips were caught by no statistical detector in 8 of 20 attack × contamination cells. Only the hash saw them. That is why the hash blocks, not the statistics.

10	**Ledger is permissioned.**	The head hash is not yet anchored to a public chain or timestamp service (future work).
11	**Variance moderation is an approximation.**	A method-of-moments empirical-Bayes prior inspired by limma, not limma itself.
12	**US-only health outcome; one-city weather.**	The Scenario Lab covers US states, 2014–2019. Other countries (for example India) have no equivalent weekly-death data wired in.
13	**Hosted deployments not tested end to end.**	Docker images and the real Render and Vercel deployments were not tested by us in this environment (their build steps were reproduced locally).
14	**Pathway-level analysis not done.**	The hallmark gene-set (GMT) file was not supplied, so pathway tests for Track A were not run.

Score: of the issues we found during the review and build, **10 groups were solved or substantially reduced** (table above), and **14 limitations remain**; of these, 1–3 and 12 depend on data we do not have or time we do not have, and the rest are accepted, documented trade-offs. None of them is hidden: each is stated in the README and on the site.

3.4 Accuracy: what it means for each part

There is no single "accuracy %" for this project, because different parts do different jobs. Using one number would be dishonest. Here are the right measures for each part.

A. The discovery engine (Track B, simulated truth, 50 seeds × 4 noise levels)

SNR = signal-to-noise ratio (higher = clearer signal). *Strict FDR* = share of reported pairs that are false (even look-alikes of true pairs). *Recall* = share of the 3 planted true interactions that were found.

	SNR 1	SNR 2	SNR 5	SNR 20
Marginal pairwise screen — strict FDR	92.0%	92.4%	92.7%	92.8%
After conditional selection — strict FDR	7.3%	0.7%	1.3%	0.7%
Full stack — strict FDR	**6.2%**	**0.7%**	**0.0%**	**0.0%**
Full stack — recall of planted interactions	**90%**	**97%**	**97%**	**97%**

Plain reading: without our protections about 9 of every 10 "discoveries" are false; with them, 0 to 6 of every 100 are false, while still finding 90–97% of the real ones.

Null calibration (100 simulations with nothing planted)	Median inflation λ	Reading
Naive test	4.11	Badly over-confident
HAC (Newey-West)	14.32	Even worse here (persistent data)
Block bootstrap (ours)	**1.38**	Much better; 7.6% of p-values below 0.05 vs ideal 5%

λ = 1 means p-values behave honestly under the null. The number comes from the median of the observed chi-square statistics divided by the median expected (0.4549).

B. Track A (real NASA genes)

- 24,688 genes tested; **0 pass FDR**; best q-value = 0.45 (needs < 0.05).
- Null calibration over 200 label permutations: λ = 0.96; 3.5% of p-values below 0.05; 6% of permutations with any hit. This shows the test is honest.
- Power by spike-in simulation: **43%** to detect a 1.5 log2 interaction; **80% only at about 4 log2**.
- Cross-tissue consistency of the exploratory top-100 genes: 55% same sign (p = 0.37, i.e. chance level).

C. Scenario model (real deaths)

Measure	Result
Training data	14,065 state-weeks, 46 states, 2014-01-04 to 2019-12-28; 200 bootstrap refits (resampling whole states)
Out-of-sample R^2 on states never seen (leave-states-out, 5 folds)	null model ≈ 0.0000; additive 0.0062; with interactions 0.0070; gradient boosting 0.0079 — i.e. **about 0.7%** of variation
Out of time (later years)	additive 0.0043; with interactions 0.0036; gradient boosting −0.0235 (the black-box model gets *worse* than guessing the average)
Calibration slope on unseen states	**0.91** (range 0.64–1.22), all causes; heart disease 0.97 (0.81–1.13); stroke 0.93 (0.60–1.21). 1.0 = perfect.
Riskiest tenth of weeks (all causes)	predicted +0.62%, observed +0.51%
Safest tenth of weeks	predicted −0.59%, observed −0.55%
Interaction check (unshrunken fit)	7 of 45 pair terms differ from zero where about 4.5 are expected by chance; an extra effect larger than about 0.37% per usual-deviation² would very likely have been seen

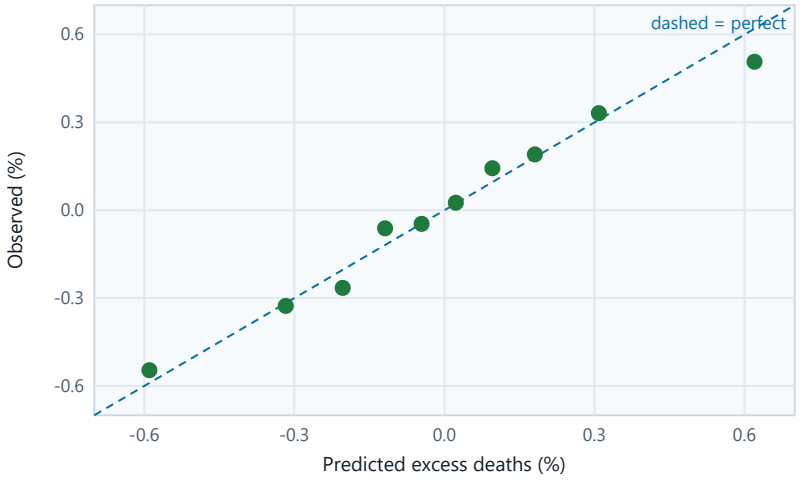


Figure: predicted vs observed excess deaths for ten equal groups of weeks the model never trained on. Points on the diagonal mean "the model's stated size is right".

> **HOW TO SAY THIS TO JUDGES**
>
> The model is **well calibrated** (when it says +0.6%, the real value is about +0.5%) but **low in explanatory power** (R^2 ≈ 0.7%), because weekly deaths are driven mostly by other things. We report both. We use the model to show direction and size of average effects with a range, never to forecast a person.

D. Security layer

- SHA-256 hash check detected **100% of 600** attack trials.
- Physical-range checks caught every decimal-shift, unit-drift and outlier injection.
- Statistical detectors: clean-data false alarms 2–5% each (isolation forest 3%, MCD 5%, autoencoder 5%, drift 2%); 10% of clean datasets trigger at least one advisory warning.
- Weak spot: subtle in-range edits (+0.3 SD) and label flips were caught by no statistical detector in 8 of 20 attack × contamination cells.

E. Software quality

85 Python tests (core, pipeline, tamper, scenario model, API); 56 frontend tests, including a Python-vs-TypeScript match to 9 decimals; three end-to-end browser suites; `make reproduce` re-runs six recorded analyses and reports REPRODUCED for all six, with 25 of 25 registered files matching the ledger; ruff lint clean; real type-check clean; `npm audit` reports 0 vulnerabilities in production dependencies.

4 Cybersecurity and the ledger

4.1 What problem does it solve?

A scientific result is the end of a long chain: raw data → cleaning → model → numbers → conclusion. An attacker, or a careless mistake, can change any link without leaving a trace: a data file edited, a pretrained model swapped, code changed after the fact, a result table altered, an old run replaced. With AI, a poisoned dataset can make a model "discover" something that is not there. The question we answer is **"how can we prove that what you are reading is exactly what the pipeline produced from exactly this data?"**

> **WHAT THE LEDGER DOES AND DOES NOT PROVE**
>
> It does **not** prove the science is right. It proves that the **computation** is tamper-evident: the data, code, settings and results are exactly what was recorded, and the analysis can be re-run to the same answer. Science still needs the statistics layer and independent replication.

4.2 What the ledger records and serves

The ledger is an **append-only, hash-chained, digitally signed log**, stored in SQLite (`artifacts/ledger.db`). Each entry is a "block". It **records**:

Block type	What it records
`GENESIS`	The first block (previous hash = 64 zeros).
`DATASET_REGISTERED`	SHA-256 of every dataset file when it enters the project (25 registered files today).
`MODEL_REGISTERED`	Hash of a model's weights, library versions, seed, training-data hash, a hold-out metric and a hash of its predictions on a fixed probe set (used to detect model swaps). Our scenario models are registered this way.
`EXPERIMENT_RUN`	The run manifest: dataset hashes, a hash of the source-code tree, configuration, seed, environment.
`FINDING_VALIDATED` / `FINDING_VERIFIED` / `FINDING_REJECTED`	The hash of a finding's scientific content (stressors, effect, p-value, n, methods, config) — never its mutable status.
`INTEGRITY_ALERT`	Any integrity failure raised by the checks.

It **serves** four things: (1) the Trust Center, which shows the chain and verifies it on demand; (2) the VERIFIED gates (G1–G4) which ask the ledger for data hashes, chain validity, finding inclusion proof and replay results; (3) the "run receipt" and Merkle proof on every finding's dossier page; (4) `make reproduce`, which compares a fresh recomputation with the ledger's fingerprints.

4.3 How a block is built (so you can explain it)

```
artifact_hashes  = list of SHA-256 hashes the block vouches for
merkle_root      = Merkle tree root of artifact_hashes  (domain-separated)
payload_hash     = SHA-256( canonical JSON of payload + artifact_hashes )
header           = { index, timestamp, block_type, prev_hash, merkle_root, payload_hash }
block_hash       = SHA-256( canonical JSON of header )
signature        = Ed25519_sign( private_key, block_hash )
```

1. **Hash chaining.** Each block contains the previous block's hash. Change one old block and every later block no longer links up.
2. **Merkle tree.** Many hashes are combined into one "root". You can prove that one specific hash is inside a block by showing a short path of sibling hashes (a **Merkle proof**) without revealing the others. That is how a finding proves it is in the ledger (gate G3).
3. **Digital signature.** The block hash is signed with an **Ed25519** private key (kept in `artifacts/keys`, never in git or Docker images). Anyone with the public key can check that the block was signed by the owner.
4. **Append-only by design.** The SQLite database has triggers that refuse any `UPDATE` or `DELETE` on the blocks table. Appends use a lock and an immediate transaction, so two writers cannot make a fork.

4.4 The five checks inside `verify()`

Error code	What was checked
`PREV_HASH_MISMATCH`	The block's previous-hash equals the real hash of the block before it.
`PAYLOAD_TAMPERED`	The stored payload re-hashes to the stored payload hash (a payload that is no longer valid JSON is also reported, not crashed on).
`MERKLE_MISMATCH`	The Merkle root recomputed from the artifact hashes matches.
`HASH_MISMATCH`	The block hash recomputed from the header matches.
`BAD_SIGNATURE`	The Ed25519 signature is valid for the block hash and signer.

4.5 The full defence layers

Layer	What it does	Blocks a result?
Data hash (SHA-256)	Compares each dataset with the hash registered in the ledger.	Yes (G1)
Ledger chain + inclusion	Verifies the whole chain; proves the finding's hash is inside a block.	Yes (G2, G3)
Reproducibility replay + code drift	Re-runs the analysis from the same seed and compares the digest of the results (floats rounded to 8 significant digits). Also compares the hash of the `src/synera_core` source tree with the one recorded at run time.	Yes (G4)

Physical-range and time-order checks	Values must be physically plausible (e.g. relative humidity 0–100, PM2.5 ≥ 0, UV 0–20) and dates strictly ordered. Skips z-scored columns that are not in physical units.	Yes (G5)
Model verification	Re-hashes the weights, re-computes the hold-out error and the hash of predictions on a fixed probe set. A swapped model fails even if the file hash were trusted.	Yes
Statistical detectors	Isolation Forest, Minimum Covariance Determinant (robust Mahalanobis), a small autoencoder, KS/PSI drift, kNN label-agreement for flipped labels, duplicates.	No — advisory only

Why the statistical detectors cannot block: they are probabilistic, and our own benchmark showed they miss subtle edits. Blocking on a guess would create false accusations; blocking on a hash mismatch is exact. This is a deliberate design decision.

How the statistical detectors stay honest (calibration)

- Each detector scores rows and flags those above the **99th percentile** of the same score on a registered **baseline** of clean data. Clean data therefore flags about 1% of rows.
- A detector raises a WARN only if the number of flagged rows is implausible under that 1% rate: a binomial test at the 0.1% level, widened by a factor of 2.5 because daily rows are autocorrelated.
- The label-flip check says "not informative" when the labels carry no structure in feature space, instead of pretending to work.
- Earlier versions fired on 100% of clean data and then tested on their own baseline rows (0% by construction). Both flaws were fixed and the false-alarm rates are now measured on held-out data.

4.6 The tamper simulator (what a judge can do)

On the Trust Center you can launch eight attacks. Each runs on a temporary copy, the real files and ledger are never touched, and the result shows which defences fired.

Attack	What it does	Who catches it
bit_flip	Flip one bit in the file.	Hash (100% of trials; SHA-256 changes in nearly all 64 characters).
subtle_edit	Nudge many values slightly, staying in range.	Hash (statistics usually miss it).
row_edit	Change a chosen cell, picking one that actually changes.	Hash.
inject_outliers	Insert extreme values.	Hash and physical-range check.
drift_units	Change units, e.g. a decimal shift.	Hash and physical-range check.
poison_labels	Flip outcome labels.	Hash (label check only if structure exists).

swap_model	Replace model weights.	Weights hash and probe-prediction hash.
ledger_tamper	Edit a block in a *copy* of the ledger.	Chain verification: valid before, invalid after; the real ledger stays valid.

The verdict shown is "RESULT BLOCKED FROM VERIFIED STATUS". In the free recorded demo, the attack outcome is a replay of a recorded attack (5% magnitude, seed 1) and the site says so.

4.7 Why not a "real blockchain"?

We need tamper evidence and an audit trail, not a public currency or distributed consensus. A signed hash chain in SQLite gives the same tamper-evidence with no gas fees, no network, instant verification and reproducible tests. The code documents the future step: anchor the head hash to a public timestamp (OpenTimestamps on Bitcoin, or a testnet contract). Be honest that today the ledger is **permissioned**: whoever holds the private key could rewrite the whole chain and re-sign it. Public anchoring would remove that last trust assumption.

5 Tech stack, models, formulas, and why

5.1 Stack at a glance

Area	Tools
Language	Python 3.11+ (science and API); TypeScript (website)
Numerics and data	NumPy, pandas, SciPy, statsmodels, patsy, pyarrow (Parquet files), PyYAML, joblib
Machine learning	LightGBM + SHAP, PyTorch (small MLP and autoencoder), scikit-learn (Isolation Forest, MinCovDet, nearest neighbours), SciPy B-splines
Provenance / crypto	hashlib SHA-256, own Merkle tree, `cryptography` library (Ed25519), SQLite with triggers
Data access	httpx, tenacity (retries), Open-Meteo, NASA POWER, CDC NCHS, EPA AQS, NASA OSDR, MyGene.info
API	FastAPI, Uvicorn, Pydantic v2, SQLModel (SQLite), Server-Sent Events
Website	React 19, TypeScript, Vite, Tailwind CSS, React Router, TanStack Query, Zustand, D3, three.js with react-three-fiber and drei, Framer Motion, Lucide icons
Quality	pytest, ruff, mypy config; Vitest, Testing Library, Playwright (end-to-end), oxlint and ESLint; GitHub-style CI configuration
Deploy	Docker Compose (api + nginx web); Vercel (static recorded demo); Render free tier (API with a low-memory guard)

5.2 The AI and ML models (deep dive)

SYNERA uses **six** modelling components. They are chosen to answer different questions, and their *disagreement* is information. The central design rule: **a flexible AI model proposes; a transparent statistical model decides.**

Model 1 — The panel interaction test (the main decision-maker)

Job: for each of the 45 stressor pairs (a, b), decide whether they interact. **Data shape:** a panel — many cities observed daily (city × day).

The model ladder (each step adds terms to the one before):

```
M0:  y ~ city fixed effects + sin/cos(day of year)
          + linear(all OTHER stressors) + spline(a) + spline(b)          (additive, smooth)
M1:  M0 + a·b                                                           (simple product
interaction)
M2:  M1 + spline(a) ⊗ spline(b)  tensor columns                          (full flexible
surface)
```

- **City fixed effects**: each city gets its own baseline, so differences between cities are not mistaken for stressor effects.
- **sin/cos of day of year**: absorbs the yearly cycle.
- **All other stressors as covariates**: stops a third stressor from posing as an interaction.
- **Cubic B-splines** (degree 3, one interior knot at the median → 5 columns per stressor): lets each stressor have a curved, not just straight, effect.
- **Joint test** M0 → M2 asks "is there any interaction at all?". Two sub-tests classify it: M0 → M1 (bilinear = SYNERGISTIC or ANTAGONISTIC) and M1 → M2 (non-linear surface = NONLINEAR).

```
F = [ (RSS0 − RSS2) / q ] / [ RSS2 / (n − k) ]     RSS = sum of squared residuals, q = terms added
```

Three p-values per test, on purpose:

- `p_naive` — the textbook F-test; assumes independent days. Shown only to display the size of the problem (λ = 4.11).
- `p_hac` — Wald test with a per-city Newey-West covariance (corrects for correlation in the errors). Sensitivity check only (λ = 14.32 here).
- `p_boot` — **the primary one**: a **block bootstrap**. Residuals of the null model M0 are resampled in moving blocks within each city (Freedman-Lane style), the F-statistic is recomputed each time (299 resamples by default), and the observed F is rescaled by the ratio of bootstrap-to-nominal medians before the F reference distribution is applied. Block length is chosen from the residual autocorrelation: twice the lag where the autocorrelation first falls inside ±2/√n, minimum 5, maximum 30 days. It is a calibration device, not an exact test (7.6% at nominal 5%).

Why this instead of alternatives:

- *vs a plain regression with an interaction term:* plain regression assumes independent errors; daily weather data are strongly autocorrelated, so it reports far too many "significant" interactions (we measured it).
- *vs only using a black-box model:* a black box gives a ranking but no p-value and no interpretable surface; it cannot be controlled for false discoveries on its own.
- *vs a mixed-effects model:* fixed effects for 8 cities are simpler, faster and sufficient; mixed models need distributional assumptions that add fragility.

Model 2 — Multiple-testing control and conditional selection

Benjamini-Hochberg false discovery rate (BH-FDR). If you test 45 pairs at p < 0.05 you expect ~2 false hits by luck alone; with 24,688 genes, over a thousand. BH controls the *share of reported findings that are false*.

```
sort p-values:  p(1) ≤ p(2) ≤ … ≤ p(m)
q(i) = min over j ≥ i of  [ p(j) · m / j ]       (monotone, capped at 1)
report if  q < α = 0.05
```

Conditional selection (forward selection + backward elimination): a pair can look significant just because it correlates with a truly interacting pair. The engine adds pairs one at a time into a combined model and

removes any whose evidence disappears once the others are in. This single step took strict false discoveries from ~92% to 0–7%.

Effect gate and confidence interval. Statistical significance is not importance. A finding must have a standardised effect of at least **0.05 of the outcome's standard deviation** and a bootstrap CI whose lower bound clears the same gate (200 resamples). The gate is pre-specified and was not tuned.

Why BH over Bonferroni: Bonferroni is too strict when many tests are related and would hide real signals; BH is the standard for discovery work (genomics) and gives an interpretable error rate.

Model 3 — LightGBM + SHAP interaction values (second opinion #1)

What it is. LightGBM is a **gradient-boosted decision tree** model: hundreds of small trees, each correcting the previous trees' mistakes. Trees automatically find thresholds and combinations without being told. **SHAP** (Shapley values) fairly splits each prediction among the features; **SHAP interaction values** split it between *pairs* of features.

```
pair strength(i,j) = mean over rows of | Φ_ij |      (symmetrised, diagonal set to 0)
```

- **Settings:** 250 trees, learning rate 0.05, 15 leaves, max depth 4, min 20 rows per leaf, row subsample 0.8, column subsample 0.9, L2 regularisation 1.0, fixed seed, single thread (repeatable).
- **Inputs:** the stressors plus nuisance columns (city code, sin/cos of day of year) so city and season are not mistaken for interaction.
- **Hold-out R^2** is computed on the last 20% of rows in time order (no shuffling), so the number is honest about overfitting.
- An optional permutation reference (refit on additive prediction + block-permuted residuals) can create a null distribution; it is off by default because it is expensive.
- **Role:** a pair counts as "supported" if it ranks in the top 6 (top-K = 6) of the 45 pairs.

Why LightGBM+SHAP: tabular data of this size is where boosted trees are strongest; SHAP is the standard way to explain them; `TreeExplainer` computes exact interaction values quickly. Versus a random forest: boosting usually fits better with fewer, shallower trees. Versus XGBoost: similar accuracy; LightGBM is faster and lighter on a laptop.

Model 4 — Neural Interaction Detection (second opinion #2)

What it is. A small PyTorch neural network (multi-layer perceptron): inputs → 32 units → 16 units → 1 output, ReLU activations. Trained with Adam (learning rate 0.003, weight decay 0.001), mini-batches of 128, up to 150 epochs, early stopping after 12 epochs without improvement on the last 20% of rows (time-ordered). Inputs and outputs are standardised. Single-threaded and seeded so the result is bit-for-bit repeatable (needed for the replay check).

How interactions are read (Tsang et al., 2018). Two inputs interact if they feed into the *same* hidden units strongly:

```
s_k          = influence of hidden unit k on the output (|path| through later layers)
strength(i,j) = Σ_k  s_k · min( |W1[k,i]| , |W1[k,j]| )
```

Why: it is a genuinely different model family from trees, so agreement between the two is meaningful. It is labelled *corroborating only* in the code: it can raise confidence but can never create a finding on its own.

Robustness score = (1 + gbm_agrees + nid_agrees) / 3. The parametric test always counts as one vote.

Model 5 — Gene-level factorial model with empirical-Bayes moderation (Track A)

Design: 2×2 (radiation yes/no × unloading yes/no) per gene on log2 expression. For each gene, ordinary least squares gives the interaction coefficient b_AB (how far the "both" group is from what adding the two single effects would predict).

```
expression ~ 1 + radiation + unloading + radiation·unloading
interaction effect  b_AB = (both − unloading) − (radiation − control)
t = b_AB / sqrt( var-scale · s̃² )
```

With 3–6 animals per group, each gene's variance estimate is very noisy. **Empirical-Bayes moderation** (Smyth 2004, the idea behind limma) shrinks each gene's variance toward the average over all genes:

```
s̃² = ( d0 · s0² + d · s² ) / ( d0 + d )       d = residual df, (d0, s0²) estimated from all genes by method of moments on log s²
```

We estimate the prior from the data (d0 ≈ 1.96 in spleen, 1.53 in skin) and compare against fixed-prior and no-moderation variants. In simulation all three keep false discoveries at 4–5%; data-estimated moderation has marginally the best power. **Why not limma itself:** limma is an R package; keeping everything in Python lets the engine, tests, replay and ledger all run in one environment. We say plainly it is an approximation.

Quality checks built in: 200 label permutations (null calibration); spike-in power simulation (known interaction added to random real genes); positive controls (overall radiation and unloading effects).

Model 6 — The Scenario model (the prediction "AI")

Job: given a state, a time of year and any mix of the 10 stressors, estimate the % change in weekly deaths and a range.

Terms (57 in total): 10 straight-line effects, heat squared (the U-shaped effect of temperature), heat × time of year and heat^2 × time of year (heat harms in summer, cold in winter), and all 45 pairwise products (possible interactions).

```
z_s        = ( x_s − normal_s(state, week) ) / usual_deviation_s(state)        (the "anomaly")
prediction = intercept + Σ β_t · term_t(z, season)
fit:   β = ( XᵀWX + λ·P·(n/100)·I )⁻¹ · XᵀW (y − ȳ)        W = weights (expected deaths), P = per-term penalty
penalty: main effects λ = 30; pairwise products λ = 300 (chosen by leave-states-out cross-validation)
```

- **Weighted ridge regression:** big states count more (weights = expected deaths). Ridge shrinks coefficients to avoid overfitting; the penalty on products is larger because interactions are the uncertain part.

- **Climatology:** each state's normal for each stressor and week is a smooth annual cycle (two sine/cosine harmonics) fitted per state.
- **Range (90% interval):** a **cluster bootstrap over states** — resample whole states, refit, 200 times. States, not weeks, are resampled because weeks inside a state are strongly dependent.
- **Extrapolation flag:** uses a Mahalanobis distance (how unusual the whole combination is compared with training data) and per-state support counts; warns when a setting is far outside what that state has experienced.
- **Second opinion:** a gradient-boosting model is trained on the same data and compared out-of-sample. It does not beat the transparent model, which justifies using the transparent one.
- **Browser twin:** the saved coefficients are evaluated by TypeScript (`frontend/src/lib/scenario.ts`), with typed arrays for the 200 bootstrap models, so the whole interval updates in ~30–115 ms. A shared test fixture proves it matches Python to 9 decimals.

Why not a deep neural network for the prediction? (1) Our signal is tiny ($R^2 \approx 0.7\%$), so a flexible model would mostly memorise noise — the boosting model already gets *worse* than guessing out of time (R^2 −0.0235). (2) We need a short list of coefficients to show "separate effects vs interaction" exactly. (3) A transparent model can be evaluated in the browser, so the free hosted site works. (4) Every number can be explained to a judge in one sentence. This is a strength to **pitch**: "we chose the simplest model that the data could support, and we proved it."

Supporting models in the security layer

Model	Role and formula
Isolation Forest	Random trees isolate odd rows faster; rows with short isolation paths score as outliers.
MinCovDet (MCD)	A robust estimate of the mean and covariance; the Mahalanobis distance $d^2 = (x-\mu)^T\Sigma^{-1}(x-\mu)$ flags rows far from the main cloud.
Autoencoder	A tiny network ($p \to p/2 \to p$, tanh, Adam, 200 epochs) that learns to rebuild normal rows; rows it rebuilds badly are unusual.
PSI drift score	$PSI = \Sigma\,(e - a) \cdot \ln(e / a)$ over 10 quantile bins of the baseline (e = expected share, a = actual share). Higher means the distribution moved.
KS test	Compares two distributions; used with PSI for drift.
kNN label agreement	Share of a row's 5 nearest neighbours with the same label; low agreement hints at flipped labels.

5.3 Validation and scoring formulas, together

Quantity	Formula / meaning
Inflation factor λ	λ = median(χ^2_1 quantile of each p-value) / 0.4549. $\lambda = 1$ means honest p-values.
Strict FDR (benchmark)	(reported pairs that are not exactly a planted pair) / (reported pairs). Look-alikes count as false.
Recall (benchmark)	(planted pairs found) / 3.

Cohen's d	(difference of means) / (pooled standard deviation). Used inside the confidence score and capped at 1.
Weighted R^2	$1 - \Sigma\, w\,(y - \hat{y})^2 \,/\, \Sigma\, w\,(y - \bar{y})^2$
Calibration slope	Slope of observed outcome regressed on the prediction; 1 = predicted sizes are right, below 1 = over-confident, above 1 = under-confident.
Confidence score (not a probability)	0.35·stat + 0.30·replication + 0.20·robustness + 0.15·trust. stat = mean of [1 − q/α (floored at 0), min(|d|, 1), 1 if the CI excludes 0]. replication = 1 independent, 0.5 same-cohort/internal, 0 none. robustness = share of agreeing methods. trust = 1 if security, chain and replay pass; otherwise **the whole score becomes 0 and the finding is BLOCKED**.
Risk-brief tier	Percentile of today's summed interaction uplift within the region's own history: LOW < 50th, MODERATE < 80th, HIGH < 95th, SEVERE otherwise; always with an uncertainty band from the finding's bootstrap CI.
Reproducibility digest	SHA-256 of the result table with floats rounded to 8 significant digits; the replay must reproduce it exactly.

5.4 Why this stack instead of the alternatives

Choice	Why we chose it	What we passed over, and why
Python for science	Best ecosystem for statistics and ML; same language for the engine, tests and replay.	R has limma, but splitting languages would break the single reproducible environment. Julia/MATLAB: smaller ecosystem for our libraries and hosting.
statsmodels + SciPy + own panel code	Full control over the exact tests (block bootstrap, QR projection) and their determinism.	Black-box AutoML would give models without interpretable p-values or false-discovery control.
LightGBM + SHAP	Strong, fast tabular model with exact tree-based interaction values.	Deep tabular models: need much more data. Random forest: weaker fit, no cheap interaction values. XGBoost: similar, heavier.
PyTorch small MLP	Lets us use a published interaction-detection method (NID) with seeded, CPU-only repeatability.	TensorFlow: heavier install. A big network: would overfit our small data.
Ridge regression for prediction	Transparent, stable with 57 correlated terms, closed-form, browser-friendly.	LASSO: unstable with correlated stressors. Deep nets/GBM: no better out of sample here and cannot be shown as separable effects.
Cluster bootstrap over states	Honest ranges when weeks in a state are dependent.	Ordinary row bootstrap or standard errors would give ranges that are too narrow.

Parquet (pyarrow)	Compact, typed, fast, easy to hash.	CSV: bigger, type-ambiguous, hash changes with formatting.
SHA-256 + Merkle + Ed25519	Standard, fast, small signatures, widely audited; Ed25519 is deterministic and has no weak-randomness pitfall.	RSA: larger, slower. A public blockchain: cost, latency, unneeded complexity.
SQLite + triggers	Zero setup, one file, transactional, enforces append-only in the database itself.	PostgreSQL: unnecessary for a single-writer ledger and not free-hosting friendly.
FastAPI	Fast, automatic `/docs` API page, type-checked requests (Pydantic), async streaming for progress.	Flask: no built-in validation or docs. Django: heavy for an API-only service.
React + TypeScript + Vite	Rich interactive UI (sliders, charts, globe) with compile-time type safety; Vite builds in seconds and outputs static files.	Streamlit/Dash: quick but limited design and interactivity, harder to host free and static. Plain JS: no type safety.
D3, three.js	D3 for custom scientific charts (contours, volcano, matrix); three.js for the globe.	Off-the-shelf chart libraries cannot draw our response surfaces and interaction maps.
TanStack Query, Zustand	Caching and loading states for API calls; tiny global store for the tour.	Redux: more boilerplate than needed.
Recorded-demo static build	Free, never sleeps, works from any computer; shows real recorded output.	Free Render server: sleeps, 512 MB RAM, cannot run heavy jobs.
Docker Compose	One command to run the full live stack identically anywhere.	Manual installs: environment drift.
Playwright, Vitest, pytest	Test real behaviour end to end (including the whole recorded demo with the API off) and numerics (Python vs TypeScript parity).	Manual clicking: not repeatable.

5.5 Datasets and licences

Data	Source	Licence
OSD-211, OSD-237 RNA-seq	NASA Open Science Data Repository / GeneLab	Public
Weather and air quality (8 cities × 518 days)	Open-Meteo archives	CC BY 4.0
Weekly deaths (all causes, heart, stroke)	CDC NCHS	Public domain
Air pollution monitors	US EPA Air Quality System	Public domain
Weather for states	Open-Meteo / ERA5	CC BY 4.0
UV index	NASA POWER	Public domain

Land outlines; Earth imagery; gene symbols	Natural Earth; NASA Visible Earth; MyGene.info	Public domain / open

5.6 Key settings (pre-specified, not tuned after seeing results)

Significance level α	0.05 (BH-FDR)
Effect gate	0.05 of outcome SD
Block-bootstrap resamples / CI resamples	299 / 200 (a quick mode uses 99 / 60)
Second-opinion top-K	6 of 45 pairs
Global random seed	20260929 (every run is repeatable)
Expression table variant	rRNA-depleted counts, chosen on technical grounds before any result was seen
Low-expression filter	mean log2 ≥ 1 and ≥ 3 samples with log2 ≥ 1; never uses group labels

6 What is left? Is anything more required?

6.1 Must do before the hackathon (checklist)

☐	Task	Why
☐	Commit and push everything, including `data/processed/us_*`, `frontend/public/snapshot/` and `data/MANIFEST.json`. Make sure no private files (`.env`, `artifacts/keys`) are committed.	The hosted demo and a clone must match what runs locally.
☐	On Render: Manual Deploy → "Clear build cache & deploy". Check `/docs` and the scenario endpoints. Vercel redeploys on push; check the tour, Scenario Lab and Findings.	Pick up the latest code and data.
☐	Dry-run the demo locally with `.\dev.cmd` (it takes 1–6 minutes per live analysis run); keep the Vercel link open as a backup.	Hosted free tier cannot run heavy computation.
☐	Rehearse with `docs/DEMO_SCRIPT.md`: start with the problem, then the tour, a live run, a tamper attack, the Scenario Lab, and the report card.	Time and confidence.
☐	Everyone on the team learns the sections of this report that match their pitch role, plus Appendix A.	Judges question individuals.
☐	Add your personal motivation story to the pitch (section 1.5).	Memorable and honest.

6.2 Nice to have (only if time remains)

- **Test Docker and the real hosted deployments end to end.** We reproduced the build steps locally but did not run the images or the deployed services ourselves.
- **Anchor the ledger head** to OpenTimestamps or a testnet — removes the "permissioned" caveat.
- **Better exposure data for Track C** (several cities per state, all monitors, population weighting) — would reduce the blur toward zero and make effects more precise.
- **An independent real dataset** for the same question (another country or period) — the only route to a VERIFIED finding.
- **An India or Punjab health dataset** if one with weekly or daily outcomes can be obtained — would make the story local.
- **Pathway-level analysis** for Track A with a hallmark GMT file.
- **Remove or retune** the MCD and autoencoder detectors (they currently add noise, not protection).
- Delete the empty `venv` and `backend\venv` folders; C: drive space is tight.

6.3 Is anything more required?

SHORT ANSWER

No new feature is required for the competition. The product is complete, tested and honest. The remaining risk is **presentation and deployment**, not capability: pushing the latest commits, redeploying, and rehearsing. Adding more features now would add risk without adding a new story; the strongest use of remaining time is a clean demo and a rehearsed answer to the hardest questions below.

A Judge questions and answers

Is the data real or simulated?

Both, and we label every number. Space biology and the US death panels are real. The environmental benchmark outcome is simulated on purpose, so we can measure our engine against a known truth. The three current validated findings are simulated and capped; none is VERIFIED.

So your engine found nothing real?

On real data it found **no interaction**, and we report that. It did find real **separate** effects: PM2.5 +0.62% deaths per +10 µg/m^3, heat in summer, cold in winter. A tool that says "no" when the data cannot support "yes" is the point of the project.

What is the "AI" in your project?

Three kinds. A **boosted-tree model (LightGBM+SHAP)** and a **neural network (NID)** propose interaction hypotheses; a **transparent ridge regression** powers the Scenario Lab; **anomaly models** (isolation forest, MCD, autoencoder) watch the data. The AI proposes; statistics decide; the ledger records.

Why not a bigger neural network?

The real signal is tiny and our out-of-sample test showed a flexible boosting model does no better, and does worse over time. A simple model we can explain, calibrate and run in the browser is the right choice for this data.

How accurate is it?

Depends which part. Engine: 90–97% recall of planted interactions with 0–6% false discoveries (simulated truth). Scenario model: calibration slope 0.91 on unseen states, but $R^2 \approx 0.7\%$, so it estimates average effects, not individual risk. Security: hash catches 100% of 600 attacks.

Does the blockchain prove your science is right?

No, and we say so on the site. It proves that the data, code and settings are exactly what was recorded and that the run replays to the same result. Scientific truth still needs independent replication.

Why not a real blockchain?

We need tamper evidence, not currency. A signed hash chain gives that with no cost or latency. Public anchoring is the documented next step.

What if someone edits one number in a dataset?

The SHA-256 changes completely, gate G1 fails, and the result is blocked. You can do this live in the Trust Center.

What if someone rewrites the ledger?

The chain of hashes and the signatures break, and verification reports exactly which block failed and why. (The remaining weakness is someone holding the private key; anchoring would fix that.)

Can I trust the Scenario Lab numbers for my city?

It covers US states, 2014–2019 (before COVID). It gives an average association with a 90% range, flags extreme settings, and states what it cannot do. It is not individual risk and not proof of cause.

Why does heat have opposite effects in summer and winter?

Hot-for-the-season weather strains people in summer; in winter, warmer-than-normal weather reduces cold stress. The model has a heat × season term because we tested it and it improved out-of-sample fit.

Why did ozone × UV disappear?

On the raw outcome, nationwide flu and weather waves created the pattern. Comparing each state only with other states in the same week removed it. We kept the honest result.

Is your deployment real?

The hosted site replays real recorded output (no server needed); the live version runs locally or in Docker and computes new runs. The site shows a "recorded demo" banner so nobody is misled.

What would make a finding VERIFIED?

Real data, an independent replication dataset, intact hashes and ledger, and an identical replay. We have the machinery; we do not yet have the second real dataset.

What did you build that is different from a standard dashboard?

The trust layer and the honesty: false-discovery control, a measured report card, a live tamper simulator, an evidence ladder that caps simulated data, and a prediction tool that shows its own limits.

B Numbers cheat-sheet

Scale	
Stressors	10
Pairs tested	45
Benchmark cities × days	8 × 518
US states (all-cause panel)	46
State-weeks (all-cause)	14,065
Years	2014–2019
Genes (spleen / skin)	24,688 / 25,716
Animals per study	21 (20 shared)
Registered files in ledger	25
Pipeline steps / gates	9 / 8 (G1–G8)
Tests	85 Python, 56 frontend, 3 end-to-end
Datasets analysed by engine	5 runnable: engine benchmark, NASA spleen (OSD-211), US all-cause, heart disease, stroke

Key results	
Marginal screen false discoveries	~92%
Full stack strict FDR	6.2% / 0.7% / 0% / 0%
Full stack recall	90% / 97% / 97% / 97%
λ naive / HAC / bootstrap	4.11 / 14.32 / 1.38
Track A genes passing FDR	0 (best q = 0.45)
Track A power	43% at 1.5 log2; 80% at ~4 log2
Real: PM2.5 per +10 $\mu g/m^3$	+0.62% (0.34–0.83)
Real: summer heat +5 °C	+0.36% (0.06–0.68)
Real: winter cold −5 °C	+0.51% (0.36–0.71)
Calibration slope (unseen states)	0.91 (0.64–1.22)
Out-of-sample R^2	≈ 0.7%
Hash detection of attacks	100% of 600
Detector false alarms (clean)	2–5% each; 10% any
Findings today	3 VALIDATED (simulated), 0 VERIFIED, 0 BLOCKED